Importação das bibliotecas

In [1]:
```
import sqlalchemy
import mysql.connector
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

Configurações

In [2]:
```
sns.set_theme(style="whitegrid")
```

Ingestão de dados

Criando a conexão com o banco de dados MySQL

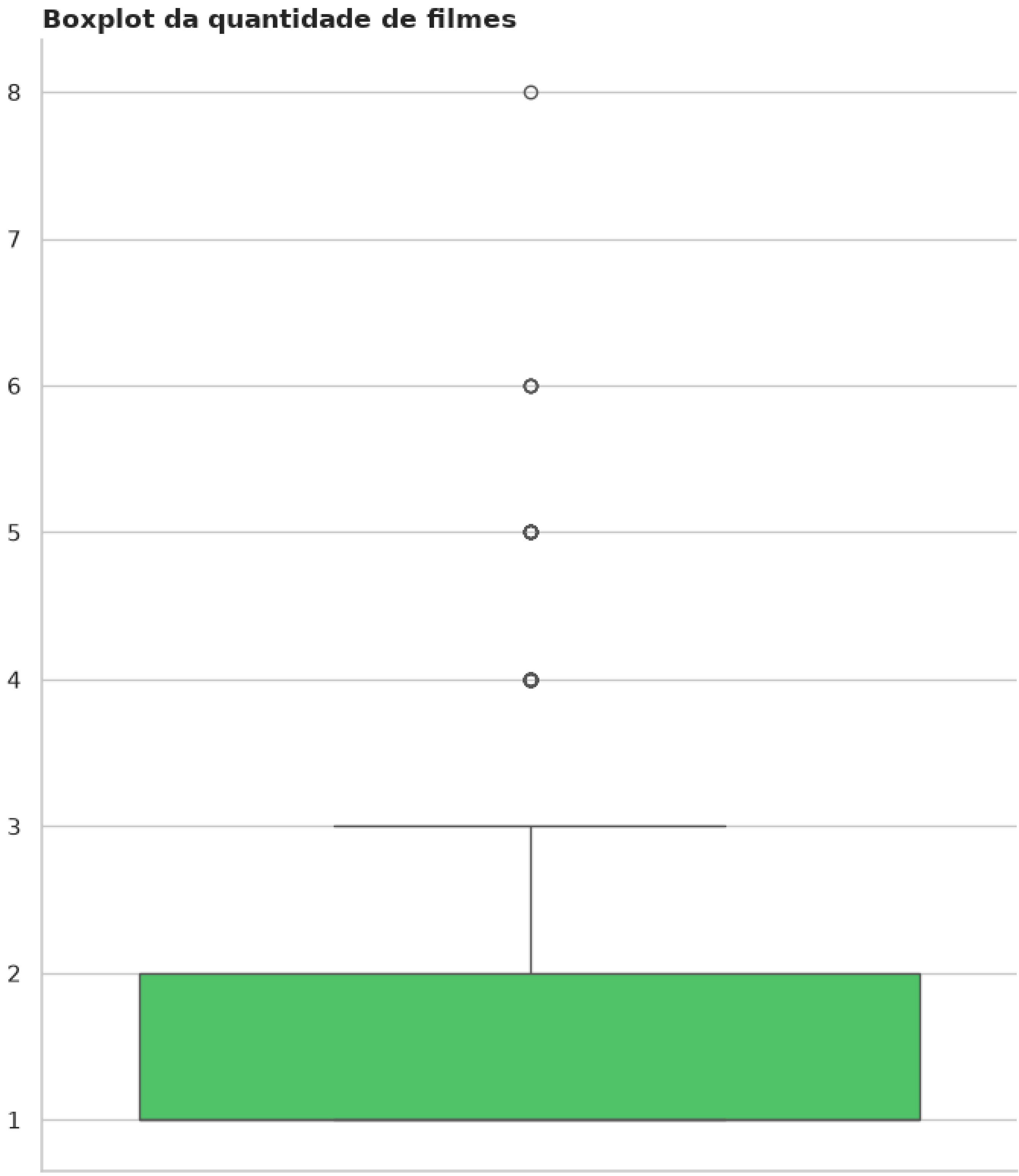
In [3]:
```
eng = sqlalchemy.create_engine('mysql+mysqlconnector://looqbox-challenge:looq-challen
```

Criando o DataFrame

In [4]:
```
query = """
SELECT *
FROM IMDB_movies
"""

df = pd.read_sql(query, con=eng)
```

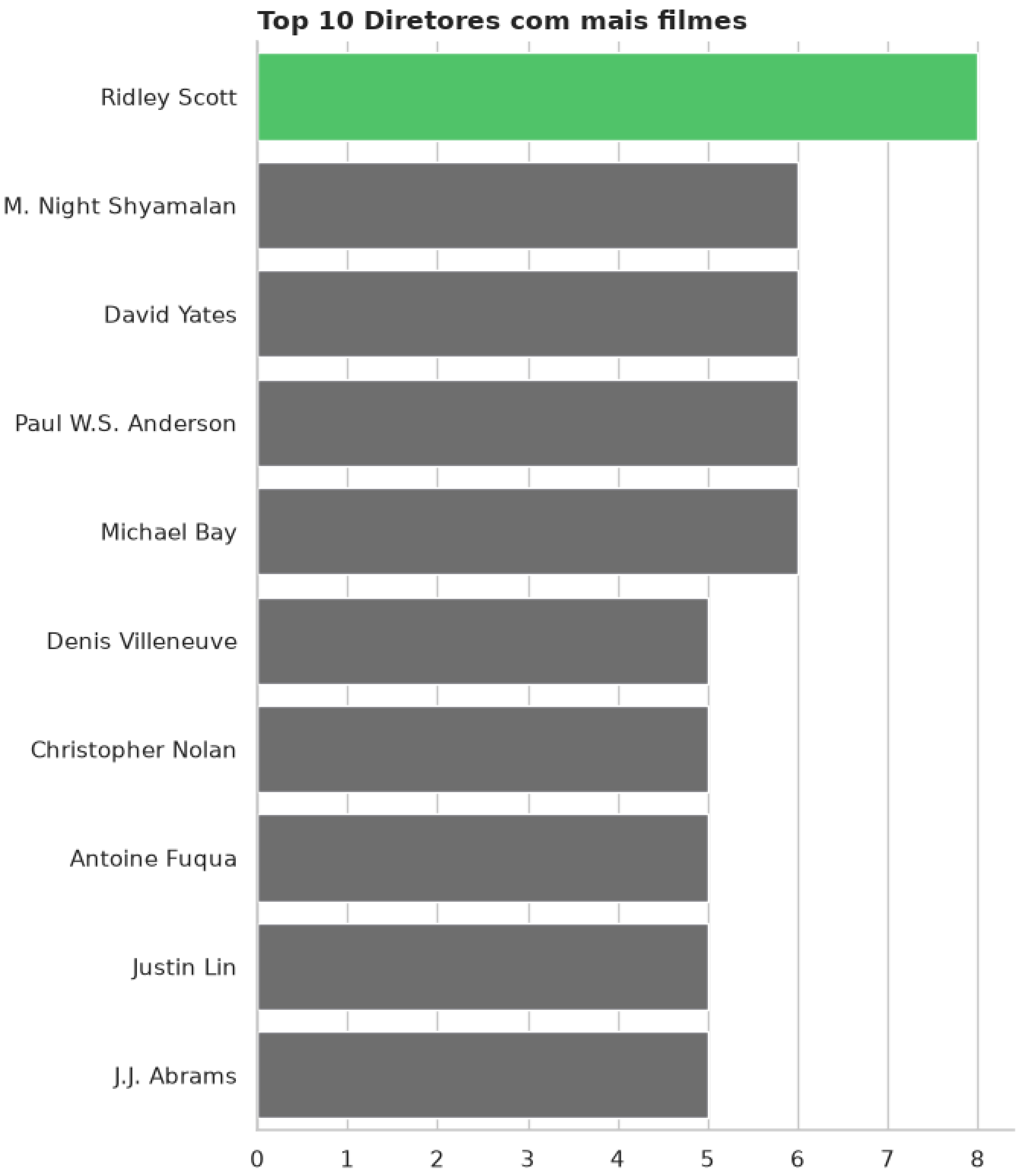
Criando imagens para analises

In [5]:
```
df_directors = (df['Director']
                 .value_counts()
                 .to_frame()
                 .reset_index())

plt.figure(figsize=(7,8))
sns.boxplot(df_directors, y='count', color='#40DA62')

plt.title('Boxplot da quantidade de filmes ',
          loc='left',
          fontdict={'fontweight': 'bold'})

plt.xlabel('')
plt.ylabel('')

sns.despine()
plt.tight_layout();
```



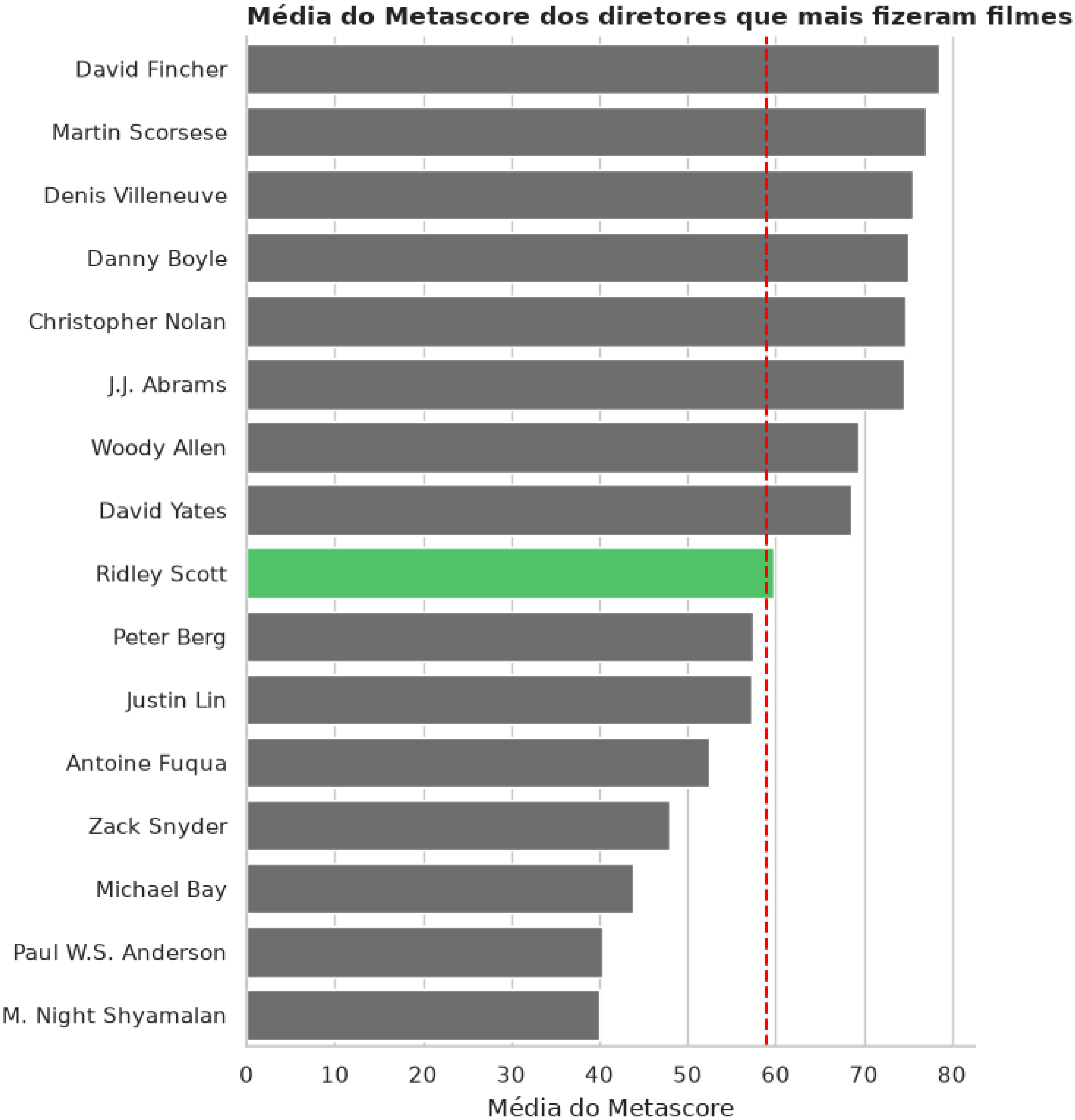
In [6]:
```
palette_ridley = [
    '#40DA62' if diretor == 'Ridley Scott' else '#6D6E71'
    for diretor in df_directors['Director'].head(10)
]

plt.figure(figsize=(7,8))
sns.barplot(data=df_directors.head(10),
            x='count',
            y='Director',
            hue='Director',
            palette=palette_ridley,
            legend=False)

plt.title('Top 10 Diretores com mais filmes', loc='left', fontdict={'fontweight': 'bo

plt.xlabel('')
plt.ylabel('')
```

```
sns.despine()
plt.tight_layout()
```



In [7]:
```
directors_above_5 = df_directors[df_directors['count'] >= 5]['Director'].to_list()
df_directors_above_5 = df[df['Director'].isin(directors_above_5)]
df_directors_mean = df_directors_above_5.groupby('Director')[['Metascore']].mean().re

metascore_mean = df['Metascore'].mean()

plt.figure(figsize=(7,8))

palette_ridley = [
    '#40DA62' if diretor == 'Ridley Scott' else '#6D6E71'
    for diretor in df_directors_mean['Director']
]

sns.barplot(data=df_directors_mean, x='Metascore', y='Director', hue='Director',palet
plt.vlines(x=metascore_mean, ymax=15.5, ymin=-0.5, colors='red', linestyles='dashed')
```

```
plt.title('Média do Metascore dos diretores que mais fizeram filmes', loc='left', fon
plt.xlabel('Média do Metascore')
plt.ylabel('')

plt.ylim(ymin=-0.5 ,ymax=15.5)

sns.despine()
plt.tight_layout()
```



In [9]:

```
df_ridley = df[df['Director'] == 'Ridley Scott']

plt.figure(figsize=(10,6))
sns.lineplot(df_ridley, x='Year', y='Metascore', color='#40DA62')
plt.hlines(y=metascore_mean, xmax=2006, xmin=2016, colors='red', linestyles='dashed')

plt.title('Nota do Metascore pelos anos de Ridley Scott', loc='left', fontdict={'font

for index, row in df_ridley.iterrows():
    plt.text(
        x=row['Year'] + 0.1,
        y=row['Metascore'],
        s=row['Title'],
        fontsize=9,
```

```
        ha='left',
        va='bottom',
        fontdict={'fontweight':700}
    )

plt.xlabel('')
plt.ylabel('')

sns.despine()
plt.tight_layout()
```